Taipei Veterans General Hospital

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## Sample Information

**Patient Name:** 周楊麗華
**Gender:** Female
**ID No.:** E201693382
**History No.:** 40540429
**Age:** 67

**Ordering Doctor:** DOC6427H 李欣恬
**Ordering REQ.:** 0CKWACE
**Signing in Date:** 2023/05/18

**Path No.:** M112-00103
**MP No.:** F23036
**Assay:** Oncomine Focus Assay
**Sample Type:** FFPE
**Block No.:** S112-16286A
**Percentage of tumor cells:** 30%

**Reporting Doctor:** DOC5466K 葉奕成 (Phone: 8#5466)

**Note:**

## Sample Cancer Type: Non-Small Cell Lung Cancer

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### Report Highlights

3 Relevant Biomarkers
9 Therapies Available
13 Clinical Trials

## Relevant Non-Small Cell Lung Cancer Variants

| Gene | Finding | Gene | Finding |
|---|---|---|---|
| ALK | None detected | NTRK1 | None detected |
| BRAF | None detected | NTRK2 | None detected |
| EGFR | ***EGFR exon 19 deletion, EGFR p.(T790M) c.2369C>T, EGFR p.(C797S) c.2389T>A*** | NTRK3 | None detected |
| ERBB2 | None detected | RET | None detected |
| KRAS | None detected | ROS1 | None detected |
| MET | None detected | | |

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.04(004).The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

## Relevant Biomarkers

| Tier | Genomic Alteration | Relevant Therapies (In this cancer type) | Relevant Therapies (In other cancer type) | Clinical Trials |
|---|---|---|---|---|
| IA | *EGFR exon 19 deletion*<br>epidermal growth factor receptor<br>Allele Frequency: 34.52% | **bevacizumab* + erlotinib** [2]<br>**erlotinib + ramucirumab** [1]<br>**gefitinib*** [2]<br>**osimertinib** [1, 2]<br>atezolizumab + bevacizumab + chemotherapy<br>gefitinib + chemotherapy | None | 12 |
| IA | *EGFR p.(T790M) c.2369C>T*<br>epidermal growth factor receptor<br>Allele Frequency: 11.25% | **osimertinib** [1, 2]<br>atezolizumab + bevacizumab + chemotherapy | None | 7 |
| IIC | *EGFR p.(C797S) c.2389T>A*<br>epidermal growth factor receptor<br>Allele Frequency: 17.81% | None | None | 4 |

**Public data sources included in relevant therapies: FDA**[1], NCCN, **EMA**[2], ESMO
**Tier Reference:** Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.
*** Includes biosimilars/generics**

**Alerts informed by public data sources:** ⊘ Contraindicated, Resistance

| | |
|---|---|
| *EGFR p.(T790M) c.2369C>T* | ⊘ **gefitinib*** [2]<br>afatinib, dacomitinib, erlotinib, gefitinib |

**Public data sources included in alerts: FDA**[1], NCCN, **EMA**[2], ESMO

### Prevalent cancer biomarkers without relevant evidence based on included data sources

*CTNNB1 p.(S37F) c.110C>T*

## Variant Details

### DNA Sequence Variants

| Gene | Amino Acid Change | Coding | Variant ID | Locus | Allele Frequency | Transcript | Variant Effect | Coverage |
|---|---|---|---|---|---|---|---|---|
| CTNNB1 | p.(S37F) | c.110C>T | COSM5662 | chr3:41266113 | 16.20% | NM_001904.4 | missense | 2000 |
| EGFR | p.(E746_A750del) | c.2235_2249delGGAATTAAGAGAAGC | COSM6223 | chr7:55242464 | 34.52% | NM_005228.5 | nonframeshift Deletion | 1944 |
| EGFR | p.(T790M) | c.2369C>T | COSM6240 | chr7:55249071 | 11.25% | NM_005228.5 | missense | 2000 |
| EGFR | p.(C797S) | c.2389T>A | COSM6493937 | chr7:55249091 | 17.81% | NM_005228.5 | missense | 1999 |
| JAK1 | p.(P733=) | c.2199A>G | . | chr1:65310489 | 99.85% | NM_002227.4 | synonymous | 1997 |
| ALK | p.(A1200=) | c.3600G>C | . | chr2:29443617 | 48.70% | NM_004304.5 | synonymous | 1963 |
| FGFR4 | p.(P136L) | c.407C>T | . | chr5:176517797 | 99.45% | NM_213647.3 | missense | 2000 |
| EGFR | p.(Q787=) | c.2361G>A | . | chr7:55249063 | 39.14% | NM_005228.5 | synonymous | 1998 |
| RET | p.(S904=) | c.2712C>G | . | chr10:43615633 | 48.80% | NM_020975.6 | synonymous | 1998 |

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## Biomarker Descriptions

**CTNNB1 (catenin beta 1)**

Background: The CTNNB1 gene encodes catenin beta-1 (β-catenin), an integral component of cadherin-based adherens junctions involved in maintaining adhesion and regulating the growth of epithelial cell layers[1]. CTNNB1 binds to the APC protein in the cytoplasm and also interacts with TCF and LEF transcription factors in the nucleus to regulate WNT signaling[2]. Steady state levels of CTNNB1 are regulated by ubiquitin-dependent proteolysis[3,4,5].

Alterations and prevalence: Recurrent somatic mutations leading to CTNNB1 activation are common in cancer. The most prevalent alterations include missense mutations in exon 3 at codons S33, S37, T41, and S45 that block phosphorylation by GSK-β and inhibit CTNNB1 degradation[6,7,8,9]. These activating mutations are observed in diverse solid tumors and have a prevalence of 20-30% in hepatocellular carcinoma, 20% of uterine carcinoma, and 15% of adrenocortical carcinoma[10,11,12,13,14,15,16].

Potential relevance: Currently, no therapies have been approved for CTNNB1 aberrations. CTNNB1 alterations have been proposed to promote cancer progression and limit the response to EGFR tyrosine kinase inhibitors in EGFR positive lung cancer[17]. Mutation of CTNNB1 is considered useful as an ancillary diagnostic biomarker for desmoid fibromatosis[18].

**EGFR (epidermal growth factor receptor)**

Background: The EGFR gene encodes the epidermal growth factor receptor (EGFR) tyrosine kinase, a member of the ERBB/human epidermal growth factor receptor (HER) family. In addition to EGFR/ERBB1/HER1, other members of the ERBB/HER family include ERBB2/HER2, ERBB3/HER3, and ERBB4/HER4[19]. EGFR ligand induced dimerization results in kinase activation and leads to stimulation of oncogenic signaling pathways including the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways. Activation of these pathways promote cell proliferation, differentiation, and survival[20,21].

Alterations and prevalence: Recurrent somatic mutations in the tyrosine kinase domain (TKD) of EGFR are observed in approximately 10-20% of lung adenocarcinoma, and at higher frequencies in never-smoker, female, and Asian populations[15,16,22,23]. The most common mutations occur near the ATP-binding pocket of the TKD and include short in-frame deletions in exon 19 (EGFR exon 19 deletion) and the L858R amino acid substitution in exon 21[24]. These mutations constitutively activate EGFR resulting in downstream signaling, and represent 80% of the EGFR mutations observed in lung cancer. A second group of less prevalent activating mutations include E709K, G719X, S768I, L861Q, and short in-frame insertion mutations in exon 20[25,26,27,28]. EGFR activating mutations in lung cancer tend to be mutually exclusive to KRAS activating mutations[29]. In contrast, a different set of recurrent activating EGFR mutations in the extracellular domain include R108K, A289V and G598V and are primarily observed in glioblastoma[24,30]. Amplification of EGFR is observed in several cancer types including 30% of glioblastoma, 12% of esophageal cancer, 10% of head and neck cancer, 5% of bladder cancer, and 5% of lung squamous cell carcinoma[15,16,23,30,31]. Deletion of exons 2-7, encoding the extracellular domain of EGFR (EGFRvIII), results in overexpression of a ligand-independent constitutively active protein and is observed in approximately 30% of glioblastoma[32,33,34].

Potential relevance: Approved first-generation EGFR tyrosine kinase inhibitors (TKIs) include erlotinib[35] (2004) and gefitinib[36] (2015), which block the activation of downstream signaling by reversible interaction with the ATP-binding site. Although initially approved for advanced lung cancer, the discovery that drug sensitivity was associated with exon 19 and exon 21 activating mutations allowed first-generation TKIs to become subsequently approved for front-line therapy in lung cancer tumors containing exon 19 or exon 21 activating mutations. Second-generation TKIs afatinib[37] (2013) and dacomitinib[38] (2018) bind EGFR and other ERBB/HER gene family members irreversibly and were subsequently approved. First- and second-generation TKIs afatinib, dacomitinib, erlotinib, and gefitinib are recommended for the treatment NSCLC harboring EGFR exon 19 insertions, exon 19 deletions, point mutations L861Q, L858R, S768I, and codon 719 mutations, whereas most EGFR exon 20 insertions, except p.A763_Y764insFQEA, confer resistance to the same therapies[39,40,41,42]. However, in 2021, the irreversible tyrosine kinase inhibitor, mobocertinib[43]was FDA approved for the treatment of NSCLC with EGFR exon 20 insertion mutations. Additionally, in 2022, the FDA granted breakthrough therapy designation to the irreversible EGFR inhibitors, CLN-081 (TPC-064)[44] and sunvozertinib[45], for locally advanced or metastatic non-small cell lung cancer harboring EGFR exon 20 insertion mutations. In lung cancer containing EGFR exon 19 or 21 activating mutations, treatment with TKIs is eventually associated with the emergence of drug resistance[46]. The primary resistance mutation that emerges following treatment with first-generation TKI is T790M, accounting for 50-60% of resistant cases[24]. Third generation TKIs were developed to maintain sensitivity in the presence of T790M. Osimertinib[47] (2015) is an irreversible inhibitor indicated for metastatic EGFR T790M positive lung cancer and for the first-line treatment of metastatic NSCLC containing EGFR exon 19 deletions or exon 21 L858R mutations. Like first-generation TKIs, treatment with osimertinib is associated with acquired resistance. In this case, resistance is associated with the C797S mutation and occurs in 22-44% of cases[46]. The T790M and C797S mutations may be each selected following sequential treatment with a first-generation TKI followed by a third-generation TKI or vice versa[48]. T790M and C797S can occur in either cis or trans allelic orientation[48]. If C797S is observed following progression after treatment with a third-generation TKI in the first-line setting, sensitivity may be retained to first-generation TKIs[48]. If C797S co-occurs in trans with T790M following sequential treatment with first- and third-generation TKIs, patients may exhibit sensitivity to combination first- and third-generation TKIs, but resistance to third-generation TKIs alone[48,49]. However, C797S occurring in cis conformation with T790M, confers resistance to first- and third-generation TKIs[48]. Fourth-generation TKIs are in development to overcome acquired C797S and T790M resistance

## Biomarker Descriptions (continued)

mutations after osimertinib treatment. EGFR targeting antibodies including cetuximab (2004), panitumumab (2006), and necitumumab (2016) are under investigation in combination with EGFR-targeting TKIs for efficacy against EGFR mutations. The bispecific antibody, amivantamab[50], targeting EGFR and MET was approved (2021) NSCLC tumors harboring EGFR exon 20 insertion mutations. The Oncoprex immunogene therapy quaratusugene ozeplasmid[51] in combination with osimertinib received a fast track designation from the FDA (2020) for NSCLC tumors harboring EGFR mutations that progressed on osimertinib alone. BDTX-189[52] was granted a fast track designation (2020) for the treatment of solid tumors harboring an EGFR exon 20 insertion mutation.

## Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

### EGFR exon 19 deletion

| Relevant Therapy | FDA | NCCN | EMA | ESMO | Clinical Trials* |
|---|---|---|---|---|---|
| osimertinib | ● | ● | ● | ● | ● (III) |
| erlotinib + ramucirumab | ● | ● | ✖ | ● | ✖ |
| bevacizumab + erlotinib | ✖ | ● | ● | ● | ✖ |
| bevacizumab (Allergan) + erlotinib | ✖ | ✖ | ● | ✖ | ✖ |
| bevacizumab (Celltrion) + erlotinib | ✖ | ✖ | ● | ✖ | ✖ |
| bevacizumab (Mabxience) + erlotinib | ✖ | ✖ | ● | ✖ | ✖ |
| bevacizumab (Pfizer) + erlotinib | ✖ | ✖ | ● | ✖ | ✖ |
| bevacizumab (Samsung Bioepis) + erlotinib | ✖ | ✖ | ● | ✖ | ✖ |
| bevacizumab (Stada) + erlotinib | ✖ | ✖ | ● | ✖ | ✖ |
| gefitinib (Mylan) | ✖ | ✖ | ● | ✖ | ✖ |
| atezolizumab + bevacizumab + carboplatin + paclitaxel | ✖ | ✖ | ✖ | ● | ✖ |
| gefitinib + carboplatin + pemetrexed | ✖ | ✖ | ✖ | ● | ✖ |
| amivantamab, lazertinib, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (III) |
| osimertinib, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (III) |
| patritumab deruxtecan | ✖ | ✖ | ✖ | ✖ | ● (III) |
| durvalumab, tremelimumab, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (II) |
| BLU-945, osimertinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |
| sunvozertinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |
| BAY-2927088 | ✖ | ✖ | ✖ | ✖ | ● (I) |
| lazertinib, amivantamab | ✖ | ✖ | ✖ | ✖ | ● (I) |
| telisotuzumab vedotin, osimertinib | ✖ | ✖ | ✖ | ✖ | ● (I) |
| TNO-155, nazartinib | ✖ | ✖ | ✖ | ✖ | ● (I) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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## Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

### EGFR p.(T790M) c.2369C>T

| Relevant Therapy | FDA | NCCN | EMA | ESMO | Clinical Trials* |
|---|---|---|---|---|---|
| osimertinib | ● | ● | ● | ● | ✖ |
| atezolizumab + bevacizumab + carboplatin + paclitaxel | ✖ | ✖ | ✖ | ● | ✖ |
| osimertinib, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (III) |
| atezolizumab, bevacizumab, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (II/III) |
| durvalumab, tremelimumab, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (II) |
| BLU-945, osimertinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |
| sunvozertinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |
| lazertinib, amivantamab, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (I) |
| telisotuzumab vedotin, osimertinib | ✖ | ✖ | ✖ | ✖ | ● (I) |

### EGFR p.(C797S) c.2389T>A

| Relevant Therapy | FDA | NCCN | EMA | ESMO | Clinical Trials* |
|---|---|---|---|---|---|
| atezolizumab, bevacizumab, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (II/III) |
| BLU-945, osimertinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |
| sunvozertinib | ✖ | ✖ | ✖ | ✖ | ● (I/II) |
| lazertinib, amivantamab, chemotherapy | ✖ | ✖ | ✖ | ✖ | ● (I) |

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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# Relevant Therapy Details

## Current FDA Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

FDA information is current as of 2023-03-15. For the most up-to-date information, search www.fda.gov.

## EGFR exon 19 deletion

### ● erlotinib + ramucirumab

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-03-22 **Variant class:** EGFR exon 19 deletion

**Indications and usage:**

CYRAMZA® is a human vascular endothelial growth factor receptor 2 (VEGFR2) antagonist indicated:

- as a single agent or in combination with paclitaxel, for treatment of advanced or metastatic gastric or gastro-esophageal junction adenocarcinoma with disease progression on or after prior fluoropyrimidine- or platinum-containing chemotherapy.
- in combination with erlotinib, for first-line treatment of metastatic non-small cell lung cancer with epidermal growth factor receptor (EGFR) exon 19 deletions or exon 21 (L858R) mutations.
- in combination with docetaxel, for treatment of metastatic non-small cell lung cancer with disease progression on or after platinum-based chemotherapy. Patients with EGFR or ALK genomic tumor aberrations should have disease progression on FDA-approved therapy for these aberrations prior to receiving CYRAMZA®.
- in combination with FOLFIRI, for the treatment of metastatic colorectal cancer with disease progression on or after prior therapy with bevacizumab, oxaliplatin, and a fluoropyrimidine.
- as a single agent, for the treatment of hepatocellular carcinoma in patients who have an alpha fetoprotein of ≥400 ng/mL and have been treated with sorafenib.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/125477s042lbl.pdf

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### ● osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-10-21 **Variant class:** EGFR exon 19 deletion

**Indications and usage:**

TAGRISSO® is a kinase inhibitor indicated for:

- as adjuvant therapy after tumor resection in adult patients with non-small cell lung cancer (NSCLC) whose tumors have epidermal growth factor receptor (EGFR) exon 19 deletions or exon 21 L858R mutations, as detected by an FDA-approved test
- the first-line treatment of adult patients with metastatic NSCLC whose tumors have EGFR exon 19 deletions or exon 21 L858R mutations, as detected by an FDA-approved test.
- the treatment of adult patients with metastatic EGFR T790M mutation-positive NSCLC, as detected by an FDA-approved test, whose disease has progressed on or after EGFR TKI therapy.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/208065s027lbl.pdf

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## EGFR p.(T790M) c.2369C>T

### osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-10-21 **Variant class:** EGFR T790M mutation

**Indications and usage:**

TAGRISSO® is a kinase inhibitor indicated for:

- as adjuvant therapy after tumor resection in adult patients with non-small cell lung cancer (NSCLC) whose tumors have epidermal growth factor receptor (EGFR) exon 19 deletions or exon 21 L858R mutations, as detected by an FDA-approved test
- the first-line treatment of adult patients with metastatic NSCLC whose tumors have EGFR exon 19 deletions or exon 21 L858R mutations, as detected by an FDA-approved test.
- the treatment of adult patients with metastatic EGFR T790M mutation-positive NSCLC, as detected by an FDA-approved test, whose disease has progressed on or after EGFR TKI therapy.

**Reference:**

https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/208065s027lbl.pdf

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## Current NCCN Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

NCCN information is current as of 2023-03-01. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

## EGFR exon 19 deletion + EGFR p.(T790M) c.2369C>T

### ● osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant classes:** EGFR T790M mutation & EGFR exon 19 deletion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Progression, Symptomatic, Asymptomatic (Subsequent therapy)

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

## EGFR exon 19 deletion

### ● osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Biomarker discovered prior to first line therapy (First-line therapy); Preferred intervention

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

### ● bevacizumab + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Non-squamous Cell; Advanced, Metastatic, Biomarker discovered prior to first line therapy (First-line therapy); Other recommended intervention
- Non-squamous Cell; Advanced, Metastatic, Biomarker discovered during first line therapy (First-line therapy)

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.04(004).

## EGFR exon 19 deletion (continued)

### erlotinib + ramucirumab

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Biomarker discovered prior to first line therapy (First-line therapy); Other recommended intervention
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Biomarker discovered during first line therapy (First-line therapy)

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

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### osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Brain Metastases, Leptomeningeal Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2022]

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### osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Stage IB , Stage IIA, Stage IIB, Stage IIIA, Stage IIIB; Resected (Adjuvant therapy)
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Biomarker discovered during first line therapy (First-line therapy); Preferred intervention
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Progression, Symptomatic, Asymptomatic (Subsequent therapy)
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Leptomeningeal Metastases, Progression (Subsequent therapy); Consider

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

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## EGFR p.(T790M) c.2369C>T

### osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**NCCN Recommendation category:** 1

**Population segment (Line of therapy):**
- Adenocarcinoma, Large Cell, Squamous Cell, Not otherwise specified (NOS); Advanced, Metastatic, Progression, Asymptomatic, Symptomatic (Subsequent therapy)

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

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### osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**NCCN Recommendation category:** 2A

**Population segment (Line of therapy):**
- Brain Metastases, Leptomeningeal Metastases (Line of therapy not specified)

**Reference:** NCCN Guidelines® - NCCN-Central Nervous System Cancers [Version 2.2022]

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## Current EMA Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

EMA information is current as of 2023-03-15. For the most up-to-date information, search www.ema.europa.eu/ema.

## EGFR exon 19 deletion

### ● bevacizumab (Allergan) + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2023-01-05 **Variant class:** EGFR exon 19 deletion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/mvasi-epar-product-information_en.pdf

### ● bevacizumab (Celltrion) + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-09-26 **Variant class:** EGFR exon 19 deletion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/vegzelma-epar-product-information_en.pdf

### ● bevacizumab (Mabxience) + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-12-15 **Variant class:** EGFR exon 19 deletion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/alymsys-epar-product-information_en.pdf

### ● bevacizumab (Pfizer) + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2023-01-05 **Variant class:** EGFR exon 19 deletion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/zirabev-epar-product-information_en.pdf

### ● bevacizumab (Samsung Bioepis) + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2023-01-05 **Variant class:** EGFR exon 19 deletion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/onbevzi-epar-product-information_en.pdf

### ● bevacizumab (Samsung Bioepis) + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-12-15 **Variant class:** EGFR exon 19 deletion

**Reference:**
https://www.ema.europa.eu/en/documents/product-information/aybintio-epar-product-information_en.pdf

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## EGFR exon 19 deletion (continued)

- **bevacizumab (Stada) + erlotinib**

  **Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2023-01-05 **Variant class:** EGFR exon 19 deletion

  **Reference:**
  https://www.ema.europa.eu/en/documents/product-information/oyavas-epar-product-information_en.pdf

- **bevacizumab + erlotinib**

  **Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-12-15 **Variant class:** EGFR exon 19 deletion

  **Reference:**
  https://www.ema.europa.eu/en/documents/product-information/abevmy-epar-product-information_en.pdf

- **bevacizumab + erlotinib**

  **Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2023-01-05 **Variant class:** EGFR exon 19 deletion

  **Reference:**
  https://www.ema.europa.eu/en/documents/product-information/avastin-epar-product-information_en.pdf

- **gefitinib (Mylan)**

  **Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-06-16 **Variant class:** EGFR exon 19 deletion

  **Reference:**
  https://www.ema.europa.eu/en/documents/product-information/gefitinib-mylan-epar-product-information_en.pdf

- **osimertinib**

  **Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-11-29 **Variant class:** EGFR exon 19 deletion

  **Reference:**
  https://www.ema.europa.eu/en/documents/product-information/tagrisso-epar-product-information_en.pdf

## EGFR p.(T790M) c.2369C>T

- **osimertinib**

  **Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-11-29 **Variant class:** EGFR T790M mutation

  **Reference:**
  https://www.ema.europa.eu/en/documents/product-information/tagrisso-epar-product-information_en.pdf

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.04(004).

## Current ESMO Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

ESMO information is current as of 2023-03-01. For the most up-to-date information, search www.esmo.org.

## EGFR exon 19 deletion

### ● osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Stage IB , Stage IIA, Stage IIB, Stage IIIA; Resected (Adjuvant therapy); ESMO-MCBS v1.1 score: A

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Early-Stage and Locally Advanced (non-metastatic) Non-Small-Cell Lung Cancer [Ann Oncol (2017) 28 (suppl 4): iv1–iv21. (eUpdate: 01 September 2021, 04 May 2020)]

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### ● osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Stage IV; Advanced, Metastatic (First-line therapy); ESMO-MCBS v1.1 score: 4

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

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### ● bevacizumab + erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**ESMO Level of Evidence/Grade of Recommendation:** I / B

**Population segment (Line of therapy):**
- Stage IV; Advanced, Metastatic (First-line therapy); ESMO-MCBS v1.1 score: 2

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

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### ● erlotinib + ramucirumab

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**ESMO Level of Evidence/Grade of Recommendation:** I / B

**Population segment (Line of therapy):**
- Stage IV; Advanced, Metastatic (First-line therapy); ESMO-MCBS v1.1 score: 3

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

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## EGFR exon 19 deletion (continued)

### gefitinib + carboplatin + pemetrexed

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**ESMO Level of Evidence/Grade of Recommendation:** I / B

**Population segment (Line of therapy):**
- Stage IV; Advanced, Metastatic (First-line therapy)

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

### atezolizumab + bevacizumab + carboplatin + paclitaxel

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**ESMO Level of Evidence/Grade of Recommendation:** III / B

**Population segment (Line of therapy):**
- Stage IV; Progression, Advanced, Metastatic (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 3

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

## EGFR p.(T790M) c.2369C>T

### osimertinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**ESMO Level of Evidence/Grade of Recommendation:** I / A

**Population segment (Line of therapy):**
- Stage IV; Advanced, Metastatic (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 4

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

### atezolizumab + bevacizumab + carboplatin + paclitaxel

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**ESMO Level of Evidence/Grade of Recommendation:** III / B

**Population segment (Line of therapy):**
- Stage IV; Progression, Advanced, Metastatic (Second-line therapy, Subsequent therapy); ESMO-MCBS v1.1 score: 3

**Reference:** ESMO Clinical Practice Guidelines - ESMO-Oncogene-addicted Metastatic Non-Small-Cell Lung Cancer [Annals of Oncology (2023), doi: https://doi.org/10.1016/j.annonc.2022.12.009 (pre-proof)]

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## Clinical Trials in Taiwan region:

## Clinical Trials Summary

### EGFR exon 19 deletion + EGFR p.(T790M) c.2369C>T

| NCT ID | Title | Phase |
|---|---|---|
| NCT04351555 | A Phase III, Randomised, Controlled, Multi-center, 3-Arm Study of Neoadjuvant Osimertinib as Monotherapy or in Combination With Chemotherapy Versus Standard of Care Chemotherapy Alone for the Treatment of Patients With Epidermal Growth Factor Receptor Mutation Positive, Resectable Non-small Cell Lung Cancer | III |
| NCT03994393 | A Phase II Trial of Durvalumab (MEDI4736) and Tremelimumab With Chemotherapy in Metastatic EGFR Mutant Non-squamous Non-small Cell Lung Cancer (NSCLC) Following Progression on EGFR Tyrosine Kinase Inhibitors (TKIs) | II |
| NCT02099058 | A Multicenter, Phase I/Ib, Open-Label, Dose-Escalation Study of ABBV-399, an Antibody Drug Conjugate, in Subjects With Advanced Solid Tumors | I |

### EGFR p.(T790M) c.2369C>T + EGFR p.(C797S) c.2389T>A

| NCT ID | Title | Phase |
|---|---|---|
| NCT04862780 | A Phase I/II Study Targeting Acquired Resistance Mechanisms in Patients With EGFR Mutant Non-Small Cell Lung Cancer. | I/II |

### EGFR exon 19 deletion

| NCT ID | Title | Phase |
|---|---|---|
| NCT04988295 | A Phase III, Open-Label, Randomized Study of Amivantamab and Lazertinib in Combination With Platinum-Based Chemotherapy Compared With Platinum-Based Chemotherapy in Patients With EGFR-Mutated Locally Advanced or Metastatic Non-Small Cell Lung Cancer After Osimertinib Failure | III |
| NCT05120349 | A Phase III, Double-blind, Randomised, Placebo-Controlled, International Study to Assess the Efficacy and Safety of Adjuvant Osimertinib Versus Placebo in Participants With EGFR Mutation-positive Stage IA2-IA3 Non-small Cell Lung Cancer, Following Complete Tumour Resection | III |
| NCT05338970 | HERTHENA–Lung02: Phase III, Randomized, Open-label Study of Patritumab Deruxtecan Versus Platinum-Based Chemotherapy in Metastatic or Locally Advanced Non-Small Cell Lung Cancer (NSCLC) With Epidermal Growth Factor Receptor (EGFRm) Mutation After Failure treatment with epidermal growth factor (EGFR) tyrosine kinase inhibitors (TKIs) | III |
| NCT03778229 | A Phase II Study Assessing the Efficacy of Osimertinib in Combination With Savolitinib in Patients With EGFRm+ and MET+, Locally Advanced or Metastatic Non Small Cell Lung Cancer Who Have Progressed Following Treatment With Osimertinib. | II |
| NCT04077463 | An Open-label Phase I/Ib Study to Evaluate the Safety and Pharmacokinetics of JNJ-73841937 (Lazertinib), a Third Generation EGFR-TKI, as Monotherapy or in Combinations With JNJ-61186372, a Human Bispecific EGFR and cMet Antibody in Participants With Advanced Non-Small Cell Lung Cancer | I |
| NCT04862780 | A Phase I/II Study Targeting Acquired Resistance Mechanisms in Patients With EGFR Mutant Non-Small Cell Lung Cancer. | I/II |
| NCT05099172 | An Open Label, First-in-human Study of BAY 2927088 in Participants With Advanced Non-small Cell Lung Cancer (NSCLC) Harboring an EGFR and/or HER2 Mutation | I |
| NCT03114319 | An Open-label, Multi-center, Phase I, Dose Finding Study of Oral TNO155 in Adult Patients With Advanced Solid Tumors. | I |

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# Clinical Trials Summary (continued)

## EGFR exon 19 deletion (continued)

| NCT ID | Title | Phase |
|---|---|---|
| NCT03974022 | A Phase I/II, Open-Label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of DZD9008 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC) with EGFR or HER2 Mutation | I/II |

## EGFR p.(T790M) c.2369C>T

| NCT ID | Title | Phase |
|---|---|---|
| NCT03178552 | A Phase II/III Multicenter Study Evaluating the Efficacy and Safety of Multiple Targeted Therapies as Treatments for Patients With Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) Harboring Actionable Somatic Mutations Detected in Blood (B-FAST: Blood-First Assay Screening Trial) | II/III |
| NCT03974022 | A Phase I/II, Open-Label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of DZD9008 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC) with EGFR or HER2 Mutation | I/II |
| NCT04077463 | An Open-label Phase I/Ib Study to Evaluate the Safety and Pharmacokinetics of JNJ-73841937 (Lazertinib), a Third Generation EGFR-TKI, as Monotherapy or in Combinations With JNJ-61186372, a Human Bispecific EGFR and cMet Antibody in Participants With Advanced Non-Small Cell Lung Cancer | I |

## EGFR p.(C797S) c.2389T>A

| NCT ID | Title | Phase |
|---|---|---|
| NCT03178552 | A Phase II/III Multicenter Study Evaluating the Efficacy and Safety of Multiple Targeted Therapies as Treatments for Patients With Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) Harboring Actionable Somatic Mutations Detected in Blood (B-FAST: Blood-First Assay Screening Trial) | II/III |
| NCT03974022 | A Phase I/II, Open-Label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of DZD9008 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC) with EGFR or HER2 Mutation | I/II |
| NCT04077463 | An Open-label Phase I/Ib Study to Evaluate the Safety and Pharmacokinetics of JNJ-73841937 (Lazertinib), a Third Generation EGFR-TKI, as Monotherapy or in Combinations With JNJ-61186372, a Human Bispecific EGFR and cMet Antibody in Participants With Advanced Non-Small Cell Lung Cancer | I |

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# Alerts Informed By Public Data Sources

## Current FDA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

FDA information is current as of 2023-03-15. For the most up-to-date information, search www.fda.gov.

## EGFR exon 19 deletion

### patritumab deruxtecan

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR exon 19 deletion

**Supporting Statement:**
The FDA has granted Breakthrough Therapy Designation to a potential first-in-class HER3 directed antibody-drug conjugate, patritumab deruxtecan, for metastatic or locally advanced, EGFR-mutant non-small cell lung cancer.

**Reference:**

https://www.cancernetwork.com/view/fda-grants-breakthrough-therapy-status-to-patritumab-deruxtecan-for-egfr-metastatic-nsclc

### osimertinib + quaratusugene ozeplasmid

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

**Supporting Statement:**
The FDA has granted Fast Track Designation to the immunogene therapy, quaratusugene ozeplasmid, in combination with EGFR inhibitor osimertinib for the treatment of non-small cell lung cancer (NSCLC) with EFGR mutations that progressed after treatment with osimertinib alone.

**Reference:**

https://www.genprex.com/news/genprex-receives-u-s-fda-fast-track-designation-for-gene-therapy-that-targets-lung-cancer/

## EGFR p.(T790M) c.2369C>T

### osimertinib + quaratusugene ozeplasmid

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

**Supporting Statement:**
The FDA has granted Fast Track Designation to the immunogene therapy, quaratusugene ozeplasmid, in combination with EGFR inhibitor osimertinib for the treatment of non-small cell lung cancer (NSCLC) with EFGR mutations that progressed after treatment with osimertinib alone.

**Reference:**

https://www.genprex.com/news/genprex-receives-u-s-fda-fast-track-designation-for-gene-therapy-that-targets-lung-cancer/

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.04(004).

## EGFR p.(C797S) c.2389T>A

### osimertinib + quaratusugene ozeplasmid

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR mutation

**Supporting Statement:**
The FDA has granted Fast Track Designation to the immunogene therapy, quaratusugene ozeplasmid, in combination with EGFR inhibitor osimertinib for the treatment of non-small cell lung cancer (NSCLC) with EFGR mutations that progressed after treatment with osimertinib alone.

**Reference:**

https://www.genprex.com/news/genprex-receives-u-s-fda-fast-track-designation-for-gene-therapy-that-targets-lung-cancer/

## Current NCCN Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

NCCN information is current as of 2023-03-01. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

## EGFR p.(T790M) c.2369C>T

### afatinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "The most common known mechanism is the acquisition of T790M (which is a secondary mutation in EGFR), which renders the kinase resistant to erlotinib, gefitinib, dacomitinib, or afatinib."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

### dacomitinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "The most common known mechanism is the acquisition of T790M (which is a secondary mutation in EGFR), which renders the kinase resistant to erlotinib, gefitinib, dacomitinib, or afatinib."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

### erlotinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "The most common known mechanism is the acquisition of T790M (which is a secondary mutation in EGFR), which renders the kinase resistant to erlotinib, gefitinib, dacomitinib, or afatinib."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

**Disclaimer:** The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.04(004).

## EGFR p.(T790M) c.2369C>T (continued)

### gefitinib

**Cancer type:** Non-Small Cell Lung Cancer **Variant class:** EGFR T790M mutation

**Summary:**
NCCN Guidelines® include the following supporting statement(s):
- "The most common known mechanism is the acquisition of T790M (which is a secondary mutation in EGFR), which renders the kinase resistant to erlotinib, gefitinib, dacomitinib, or afatinib."

**Reference:** NCCN Guidelines® - NCCN-Non-Small Cell Lung Cancer [Version 2.2023]

## Current EMA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

EMA information is current as of 2023-03-15. For the most up-to-date information, search www.ema.europa.eu/ema.

## EGFR p.(T790M) c.2369C>T

### gefitinib

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-07-05 **Variant class:** EGFR T790M mutation

**Reference:**

https://www.ema.europa.eu/en/documents/product-information/iressa-epar-product-information_en.pdf

### gefitinib (Mylan)

**Cancer type:** Non-Small Cell Lung Cancer **Label as of:** 2022-06-16 **Variant class:** EGFR T790M mutation

**Reference:**

https://www.ema.europa.eu/en/documents/product-information/gefitinib-mylan-epar-product-information_en.pdf

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